

# BERNARDO GUTERRES

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<https://bernardoguterres.github.io/portfolio/>

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## EDUCATION

### Bachelor of Science in Computer Science

Sep 2023 - June 2026

King's College London

- Third Year Student | First-Class Average: 77%
- Societies: KCL Tech, KCL Volleyball, KCL Football

### International Baccalaureate

Sep 2021 - July 2023

St Dominics International School, Portugal

- 38/45 Points Overall | Physics HL: 7, Economics HL: 7, Mathematics AA HL: 6

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## WORK EXPERIENCE

### Software Engineer Intern, Link Consulting

Jun 2025 - Aug 2025

- Engineered an AI-powered database migration platform (LACE PLSQL Copilot) integrating Next.js, React, and Claude API to automate PL/SQL to Spark SQL conversion, reducing manual refactoring time
- Built full-stack web application featuring real-time data streaming, smart context loading, and interactive visualizations, improving client engagement with migration progress
- Implemented dual-backend architecture (Node.js, Flask) enabling parallel processing of database modernization workflows

### Analyst(Part Time), Minds of Tomorrow

Jun 2025 - Present

- Authored 8+ monthly analytical articles on diverse trends (remote work, tariffs) for a multidisciplinary content platform connecting 600+ students and graduates across professional fields

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## PROJECTS

### AlphaLab

Feb 2026 - Present

- Built a desktop algorithmic trading backtesting platform using a React/TypeScript frontend, Flask REST API backend, and Tauri (Rust) for native macOS/Windows/Linux packaging at under 10MB
- Implemented realistic execution simulation with configurable slippage, commissions, and next-bar execution (no look-ahead bias), alongside 50+ technical indicators and 30+ performance metrics including Sharpe, Sortino, and VaR
- Engineered walk-forward validation and Monte Carlo simulation to detect overfitting and assess outcome distributions across 5 built-in trading strategies
- Designed JSON strategy export system enabling seamless integration with AlphaLive for live trading deployment, with 120+ passing tests across backend and frontend

### AlphaLive

Feb 2026 - Present

- Built a 24/7 live algorithmic trading execution engine deployed on Railway, connecting to Alpaca Markets broker API to automatically generate signals and execute trades from strategies exported by AlphaLab
- Implemented production-grade risk management including stop loss, take profit, trailing stops, position sizing, daily loss limits, and circuit breakers (consecutive loss detection, kill switch, corporate action detection)
- Integrated real-time Telegram notifications for trade alerts, position closes, and daily P&L summaries, with graceful degradation ensuring trading continues if alerts fail

### NBA MVP Predictor

May 2025

- Created predictive model forecasting NBA MVP vote shares using Ridge Regression, Random Forest, and XGBoost trained on 33+ years of Basketball-Reference.com data (15,000+ player-seasons)
- Engineered 52 features across 7 categories including era-normalised ratio features, league-relative z-scores, and narrative/history features to mirror real voter behaviour patterns
- Implemented time-aware rolling backtest (train on all years < Y, test on Y) to prevent data leakage, validated with ranking-focused metrics including MRR, Precision@5, and Top-3 Recall
- Applied SHAP analysis identifying free throw attempts and field goals as top predictive features, with a 71-test suite covering feature engineering, data validation, and backtest integrity

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## ADDITIONAL INFORMATION

- **Technical Skills:** Python, JavaScript, React.js, Next.js, Machine Learning, SQL, Rust, C++
- **Languages:** Portuguese: Native | English: Fluent | Spanish: Proficient | French: Conversational
- **Certifications:** Data Scientist Certificate, BCG Data Science Job Simulation, Machine Learning Specialization